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A genetic algorithm optimized hybrid model for agricultural price forecasting based on VMD and LSTM network

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Accurately predicting agricultural commodity prices is challenging due to their unpredictable and complex nature. Existing models often fail to capture nonlinear and nonstationary patterns in price data, resulting in less accurate forecasts. To tackle these challenges, we present a novel hybrid VMD-LSTM model that synergistically combines genetic algorithm (GA), variational mode decomposition (VMD), and long short-term memory (LSTM), leading to better prediction accuracy. The proposed model utilizes GA-optimized VMD, which decomposes a price series into intrinsic mode functions (IMFs) with a unique property of sparsity leading to faster convergence. Then, these IMFs are individually modelled and forecasted using GA-optimized LSTM models. Finally, the forecasts of all IMFs are ensembled to provide an output for the actual price series. VMD-LSTM is evaluated against individual LSTM and decomposition-based models (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM) using monthly price data for maize, palm oil, and soybean oil. Performance is assessed through root mean square error (RMSE), mean absolute percentage error (MAPE), and directional prediction statistics (D_{stat}). VMD-LSTM reduces RMSE by 56.93%, 21.83%, and 27.00% and MAPE by 44%, 21.67%, and 25.85% for maize, palm oil, and soybean oil, respectively, compared to the next best CEEMDAN-LSTM. TOPSIS and Diebold-Mariano test also confirm the better prediction accuracy of VMD-LSTM. The proposed model can be a better tool for agricultural price forecasting, supporting decision-making for farmers, traders, and policymakers.

Keywords Agricultural price, Genetic algorithm, Hybrid model, Intrinsic mode functions, Long short-term memory, Variational mode decomposition

Agricultural commodity prices are crucial in influencing both producers' motivation to cultivate and consumers' ability to afford food. This presents a challenging situation for policymakers who must navigate the competing interests involved. Reliable forecasts of agricultural commodity prices are vital for effective marketing strategies, investment choices, and the planning of agricultural policies[1]. The agricultural commodity market is shaped by a wide range of interconnected factors. These include fluctuations in climate, which can significantly affect crop yields; seasonal patterns in production, where certain crops are grown and harvested during specific times of the year; and derived demand, which refers to the demand for agricultural products driven by their use in other industries. Additionally, imperfections in the market, such as lack of transparency or unequal access to resources, play a role. Economic globalization further adds complexity by exposing the market to international trade dynamics and competition. Lastly, administrative regulations, such as government policies and trade restrictions, have a direct impact on market operations and pricing. These factors make the price series extremely complex, nonlinear, nonstationary, and chaotic, making price prediction an extremely challenging task[2]. Accurate and reliable forecasts of agricultural commodity prices are critical for balancing the interests of producers and consumers, promoting market stability, and enabling informed decision-making. In this regard, the research community has put enormous efforts in order to improve the accuracy of price forecasting of agricultural commodities. These efforts lead to the generation of several novel techniques for time series modelling[3]. Among these models, artificial intelligence (AI) techniques, particularly artificial neural networks (ANNs), have garnered considerable interest owing to their potent self-learning capabilities and adaptable

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architectures[4,5]. Neural network models have a notable advantage over other nonlinear models i.e. the ability to accurately approximate any continuous function when trained with a sufficient dataset[6]. This feature has made neural networks widely utilized in agricultural price prediction. For instance, Jha and Sinha (2014)[7] employed a time-delay neural network (TDNN) to forecast various oilseed crops' monthly wholesale price series. The study compared the TDNN model's performance with the autoregressive integrated moving average (ARIMA) model and found that the TDNN model exhibited superior performance for nonlinear series. Even though the ANNs have gained much popularity for their wide use in real world scenario, they have several limitations due to their shallow buildup[8]. Among these limitations, some include lack of depth in architecture, suffering from local optima problems, overfitting and requirement of expert for feature selection. These limitations are addressed by the more advanced technique called deep learning (DL). DL is a subset of machine learning which have depth in layers and complex architectures that can captures the features as well as do the task of modelling, classification and prediction by own. DL approaches improve traditional ANNs by enhancing their complexity and precision, leading to successful applications in various fields[9]. An increasingly promising deep learning technique for price forecasting is the long short-term memory (LSTM) model, which excels at learning temporal sequences and is well-suited. LSTM belongs to the family of recurrent neural networks and employs sophisticated gated mechanisms to retain information over extended periods, effectively addressing the vanishing gradient problem[10]. It also handles long-term dependencies, which are crucial in price forecasting. Overall, the potential of DL methods, particularly LSTM, in time series forecasting offers exciting prospects for improving accuracy and precision in agricultural commodity markets. Jaiswal et al. (2021)[11] demonstrate superior forecasting performance of the LSTM model over ANN and ARIMA models for international palm oil and maize prices.

Considering the limitations of an individual model for complex agriculture price series, hybrid models have been preferred in the literature[12]. There are broadly two types of hybrid modelling, one is combination of two or more forecasting models reaping the benefits of each, while the other is decomposition based hybrid models in which original series is decomposed into several uniform or simpler subseries with same length as of the original series and then forecasting of each subseries with any state of the art model. The later one has better results and thus utilized in this study[13–15]. The empirical mode decomposition (EMD) method has gained widespread popularity for decomposing nonlinear and nonstationary data[16]. For instance, Yu et al. (2008)[17] applied EMD in world crude spot price forecasting using an EMD-based ANN model. Similarly, Lin et al. (2012)[18] utilized an EMD-based hybrid model for foreign exchange rate forecasting, combining least squares support vector regression (LSSVR) with EMD. However, EMD is also suffering with a major limitation called model mixing and this leads to the improvisation of this technique which gave improved variants of EMD like ensemble empirical mode decomposition (EEMD)[19] and complementary ensemble empirical mode decomposition with adaptive noise (CEEMDAN)[20]. For example, Choudhary et al. (2019)[21] used EEMD for agriculture commodity price analysis. Fang et al. (2020)[22] utilized EEMD to decompose various agricultural commodities and then employed ARIMA, ANN, and support vector machine (SVM) models to predict the decomposed components. Zhang et al. (2014)[23] proposed an EEMD-based hybrid model that integrated generalized autoregressive conditional heteroscedasticity (GARCH) and LSSVM to forecast crude oil prices. Prasad et al. (2018)[24] showcased the superiority of a hybrid model that combines CEEMDAN and an extreme learning machine (ELM) for soil moisture forecasting. Cao et al. (2019)[25] combined CEEMDAN with LSTM for stock market prices forecasting. Meanwhile EMD and its variants have been applied successfully in some time series forecasting applications, but still have many limitations[26–28]. One major challenge of these methods is the issue of boundary effects, which occur when the decomposition process leads to the creation of artificial extrema at the boundaries of the data, thus introducing errors into the final prediction results. In addition, EMD and its variants are prone to mode mixing, where the resulting decomposed components may contain mixed information from multiple underlying modes, leading to inaccurate predictions. These methods are also highly sensitive to noise, which can have a significant impact on the decomposed components and the resulting forecasts. Finally, while EMD and its variants are widely used in practice, they lack exact mathematical foundations, making it difficult to interpret their results and compare them with other methods. These limitations could potentially impact the accuracy and reliability of the final prediction results.

Variational mode decomposition (VMD) is a new decomposition technique that has been introduced to address the limitations of EMD variants[29]. VMD is a non-recursive and data-adaptive method that decomposes a time series into modes with highly uncorrelated frequency content. The advantages of VMD over EMD include reduced boundary effects, improved mode separation, reduced sensitivity to noise, and a more precise mathematical foundation. As such, VMD is becoming an increasingly popular technique for time series analysis and forecasting. The use of VMD in conjunction with the forecasting model has been found to produce superior results in stock price forecasting[30–32], carbon price forecasting[33], and non-ferrous metals price forecasting[34]. They concluded that forecasting results obtained from VMD-based hybrid models are more accurate and reliable than those obtained using traditional EMD-based hybrid models. In recent years, various studies have been conducted to develop hybrid models based on combining VMD and LSTM[31,35,36]. Despite the advantages of VMD and LSTM, the accuracy of these hybrid models remains unsatisfactory due to the lack of standard rules for selecting the hyperparameters of both tools. Therefore, optimization techniques such as genetic algorithm (GA) have gained attention in recent studies to determine hyperparameters' optimal values[37]. The GA is a powerful optimization technique that can search the vast solution space of hyperparameters and find the best combination of hyperparameters to minimize the prediction error of the hybrid model. Using GA, the hybrid models can capture the sharp fluctuations in the agricultural price series and provide more accurate and reliable forecasts for agricultural stakeholders.

This paper proposes a novel GA-optimized hybrid model, VMD-LSTM, that harnesses the unique advantages of GA, VMD, and LSTM techniques, each contributing to optimization, decomposition, and prediction capabilities, respectively. Our model is designed to achieve superior prediction accuracy compared to existing

individual and hybrid models. The methodology of our proposed model comprises three key steps that set it apart from existing approaches. Firstly, we employ GA-optimized VMD to decompose the complex nonlinear and nonstationary price series into a set of IMFs. By incorporating GA optimization, this decomposition step aims to minimize the loss and enhance the effectiveness of the process, surpassing existing EMD variants. The second step involves individual forecasting of each decomposed IMF using GA-optimized LSTM. By optimizing the parameters of the LSTM model through GA, we further improve the accuracy of the predictions, leveraging the powerful learning capabilities of LSTM. Finally, we ensemble the predicted values of each IMF to obtain a comprehensive and refined prediction result for the complex agricultural price series. This novel hybrid approach combines the strengths of GA, VMD, and LSTM, offering a comprehensive and effective solution for accurately forecasting agriculture commodity prices. The paper's most significant contributions are outlined as follows:-.

1) We proposed a novel hybrid model for agricultural price forecasting that combines the optimization capabilities of GA, the decomposition power of VMD, and the prediction strength of LSTM. Here, GA-optimized VMD disintegrates a highly complex agricultural price series into a set of IMFs with minimum decomposition loss compared to existing EMD variants. Further, for modelling each IMF, we utilized GA-optimized LSTM, aiming to enhance the overall prediction accuracy of our hybrid model.
2) We use GA as an optimization technique to solve the parameter selection problem for VMD and LSTM.
3) We use three real agricultural price series to evaluate our proposed models empirically. The results demonstrate the superior performance of our proposed hybrid model compared to baseline models concerning both level and direction prediction accuracy. To comprehensively understand and rank all the models, we employed a revolutionary technique called TOPSIS (technique for order of preference by similarity to ideal solution), which effectively ranks each model by combining their level and directional performances.
4) We developed and published two packages of R software namely, *decompDl*[38] and TSdeeplearning[39], in CRAN for practical use.

The subsequent sections of the paper are structured as follows: The methodology section elaborates the proposed hybrid model. Results and discussion section discusses the empirical results of applying the proposed hybrid and the baseline models to three international agriculture price series, followed by the conclusions for this paper.

Methodology

Genetic algorithm optimized variational mode decomposition

VMD[29] is a novel data-adaptive decomposition technique that has emerged as a promising decomposition to overcome many limitations of EMD variants[19,40]. The VMD algorithm extracts modes concurrently in a non-recursive manner, assuming a limited bandwidth of the central frequency for each intrinsic mode function (IMF). VMD decomposition yields modes with a unique property of sparsity, where each mode is mainly centred around a specific pulsation in the frequency spectrum. This characteristic distinguishes VMD from other decomposition techniques. Additionally, VMD offers several advantages, making it superior in handling complex data, particularly nonstationary and nonlinear data. Primarily, the modes are robust to noise, essential when dealing with complex agricultural price data consisting of multi-frequency series.

Additionally, VMD has faster convergence and better accuracy. These characteristics of VMD make it highly suitable for addressing complex agricultural price data consisting of multi-frequency signals. The following steps are employed to estimate the bandwidth of a mode:

1) The Hilbert transformation (HT) is applied to each mode $c_j\,(t)$ to obtain a one-sided frequency spectrum.
2) the frequency spectrum of the mode is shifted by employing the modulation properties
3) Finally, the bandwidth of $c_j\,(t)$ is estimated using H^1 Gaussian smoothness of the demodulated signal, which corresponds to squared L^2-norm of the gradient.

The following constraints of the variation problem can be used to explain VMD:

$$\min_{\{\omega_j\},\{c_j\}} \left\{ \sum_j \parallel \partial_t \left[\left(\delta\,(t) + \frac{i}{\pi\, t} \right) * c_j\,(t) \right] e^{-i\omega_j t} \parallel_2^2 \right\}$$

$$s.t. \sum_{j=1}^{n} c_j\,(t) = y\,(t)$$

where $y\,(t)$ is the original price series, $\{c_j\} := \{c_1, c_2, \cdots c_n\}$ is the set of modes, $\{w_j\} := \{w_1, w_2, \cdots w_n\}$ is the of central frequencies, $\delta\,(t)$ is the impulse function, $\partial_t(.)$ is the partial derivative of time t, n is the number of modes, * denotes convolution operation, $\parallel . \parallel$ denotes norm processing, and $i = \sqrt{-1}$. Lagrangian multipliers $\lambda\,(t)$ and quadratic penalty terms are used to transform a constraint problem into an unconstrained problem that is easy to solve:

$$L\left(\{c_j\},\{\omega_j\},\lambda\right) = \alpha \sum\nolimits_j \parallel \partial_t \left[\left(\delta\,(t) + \frac{i}{\pi\, t} \right) c_j\,(t) \right] e^{-i\omega_j t} \parallel_2^2 + \parallel y\,(t) - \sum_j c_j\,(t) \parallel_2^2 + \left[\lambda\,(t), y\,(t) - \sum_j c_j\,(t) \right]$$

where α is said to be a balance parameter or penalty parameter of data fidelity constraint.

Furthermore, an iterative sequence called the alternate direction method of multipliers (ADMM) is applied to obtain the saddle point of the above equation. Finally, VMD requires four hyperparameters i.e. number of modes ($\boldsymbol{n}$), balancing parameter of the data-fidelity constraint ($\boldsymbol{\alpha}$), tolerance of convergence criterion ($\boldsymbol{\tau}$), and time-step of the dual ascent ($\boldsymbol{\epsilon}$), which impact the decomposition's architecture, and selecting the optimum values of these parameters is crucial for a given data set. To obtain these hyperparameters with minimum decomposition loss, GA-based VMD[31] is used for better decomposition performance. Here is a brief overview of how genetic algorithm-based VMD works:

1) initialization: the algorithm starts by generating an initial population of VMD parameters
2) Evaluation: Each set of VMD parameters decomposes the time series into its constituent modes (IMFs). After taking the initial parameter, VMD's decomposed IMFs follow as: $\acute{y}(t) = \sum_{j=1}^{n} c_j(t)$. Ideally, the sum of all IMFs should be the same as the original series as: $\acute{y}(t) = y(t)$. However, it does not generally hold, as the VMD causes decomposition loss. Hence, we define the fitness function which minimizes decomposition loss, i.e. the ratio of residual energy $(r_{res}) = \frac{\sum_{t=1}^{T} \left| \frac{y(t) - \sum_{j=1}^{c_j(t)}}{y(t)} \right|}{T}$, where T is the total observations.
3) Selection: The algorithm selects the best-fit sets of VMD parameters from the current population to be parents for the next generation. The selection process is based on the fitness function values of the solutions. Table 1 shows the hyperparameter range for GA-optimized VMD.
4) Crossover: The selected parents are combined through a crossover operator to create offspring solutions that inherit some characteristics from each parent.
5) Mutation: Some offspring solutions undergo a mutation operator that randomly changes some VMD parameters to introduce new variations in the population. the mutation operator plays a crucial role in maintaining the diversity of the population and preventing the algorithm from converging prematurely to local optima. By introducing random changes to the individuals (solutions) in the population, the mutation operator ensures that the search process does not become overly focused on a specific region of the solution space. This randomness allows the algorithm to explore new and potentially better regions of the search space, improving the chances of finding a global optimum or a more optimal solution. In essence, mutation adds variability and adaptability, which are vital for the effectiveness and robustness of genetic algorithms.
6) Replacement: In this step, the newly generated offspring solutions are introduced into the population by replacing the least-fit solutions. This process ensures that the weaker individuals, which are less likely to contribute to an optimal solution, are removed. As a result, a new generation of VMD parameters is created, gradually steering the population towards better solutions.
7) Termination: the algorithm stops when a termination criterion is met.

Once all hyperparameters are optimized, the obtained IMFs are modelled individually through the LSTM model to help improve the prediction accuracy of an agriculture price series.

Genetic algorithm optimized long short-term memory model

LSTM[41] is a type of RNN architecture well-suited for processing sequential data. LSTMs are specifically designed to address the challenge of vanishing gradients, a common issue faced by traditional RNNs when processing long sequences. LSTMs achieve this by introducing memory cells that can store and retain information for extended durations. LSTMs utilize memory cells to capture long-term dependencies in sequential data effectively. The LSTM architecture consists of three fundamental gates: the forget gate (f_t), the input gate (i_t), and the output gate (o_t). Alongside these gates, the cell state (C_t) plays a crucial role in the LSTM model. It spans across the entire sequence, running in parallel, and facilitates minor interactions. The initial step in information processing with LSTMs involves discarding irrelevant information from the cell state, thereby enabling the model to focus on relevant features and patterns in the data. To achieve this, the first gate, referred to as the "forget gate," employs a sigmoid function expressed in the following equation:

$$f_t = \sigma\left(W_f \bullet [h_{t-1},\ y_t] + b_f\right)$$

The subsequent step involves storing new or relevant information in the cell state. This is accomplished through another gate that combines a sigmoid function known as the "input gate" and a tanh layer, as expressed in the following equations:

Hyperparameters	Range
Number of modes ($\boldsymbol{n}$)	$\{2,3,4 \cdots 25\}$
Balancing parameter of the data-fidelity constrain ($\boldsymbol{\alpha}$)	$\left\{10, 10^2,\ 10^3, 10^4,\ 10^5\right\}$
Tolerance of convergence criterion ($\boldsymbol{\tau}$)	$\{0,1\}$
Time-step of the dual ascent: ($\boldsymbol{\epsilon}$)	$\left\{10^{-4},\ 10^{-5}, 10^{-6},\ 10^{-7}\right\}$

Table 1. Range of hyperparameters for genetic Algorithm-Optimized variational mode decomposition.

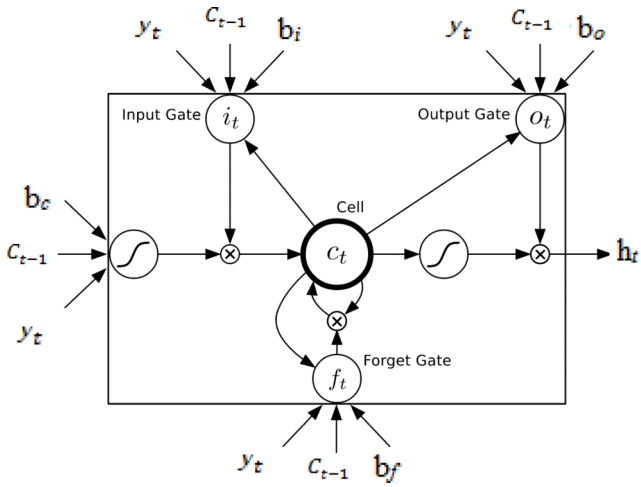


Fig. 1. Long short-term memory cell.

Hyperparameters	Range
Input node	$\{1,2,3\cdots 15\}$
Hidden nodes	$\{1,2,3\cdots 15\}$
Batch size	$\{20,\ 25\cdots 70\}$
Number of epochs	$\{30,\ 40\cdots 100\}$

Table 2. Hyperparameters range for genetic algorithm optimized long short-term memory model.

$$i_t = \sigma\left(W_i \bullet \left[h_{t-1},\ y_t\right] + b_i\right)$$

$$\widetilde{C}_t = tanh(W_c \bullet \left[h_{t-1},\ y_t\right] + b_c)$$

Lastly, the output is determined based on the updated cell state $\widetilde{C}_t$, utilizing a sigmoid layer and a *tanh* layer, which selectively output the relevant parts as expressed in the following equations:

$$C_t = f_t \times C_{t-1} + i_t \times \widetilde{C}_t$$

$$o_t = \sigma\left(W_o \bullet \left[h_{t-1},\ y_t\right] + b_o\right)$$

$$h_t = o_t \times \tanh\left(C_t\right)$$

The equations mentioned above involve weight matrices ***W***, a bias term (b), sigmoid function (σ (.)), and hyperbolic tangent function (*tanh*(.)). Using these equations, Fig. 1 illustrates how information flows within a single LSTM cell. The training of LSTM model requires the fine tuning of several hyperparameters which is to be set prior to running on the dataset. These hyperparameters include the numbers of input nodes, hidden nodes, hidden layers, epochs and batch size. During our observations, we found that the number of hidden nodes and inputs significantly influenced the model's performance, while other factors, such as the number of epochs, batch size, and the type of activation function, had minimal effects. Finding the optimal set of hyperparameters is often challenging, as it requires a lot of trial and error, and the performance of the LSTM can be highly sensitive to even small changes in the hyperparameters. Optimizing the hyperparameters can help improve the accuracy and generalization ability of the LSTM, leading to better prediction performance of individual IMF. GA-based optimization can help overcome the limitations of manual tuning of significant hyperparameters[42]. It can also help to avoid overfitting by selecting the optimal combination of hyperparameters.

The genetic algorithm operates by generating an initial set of possible solutions. This population of solutions is then iteratively improved through a process that mimics natural selection, where the fittest solutions are selected and recombined to create new, potentially better solutions in subsequent generations. Each individual in the population represents a set of hyperparameters for the LSTM model. The range of these hyperparameters is shown in Table 2. Overall, a genetic algorithm-based LSTM model can be a powerful tool for optimizing the performance of LSTM models on individual IMF prediction.

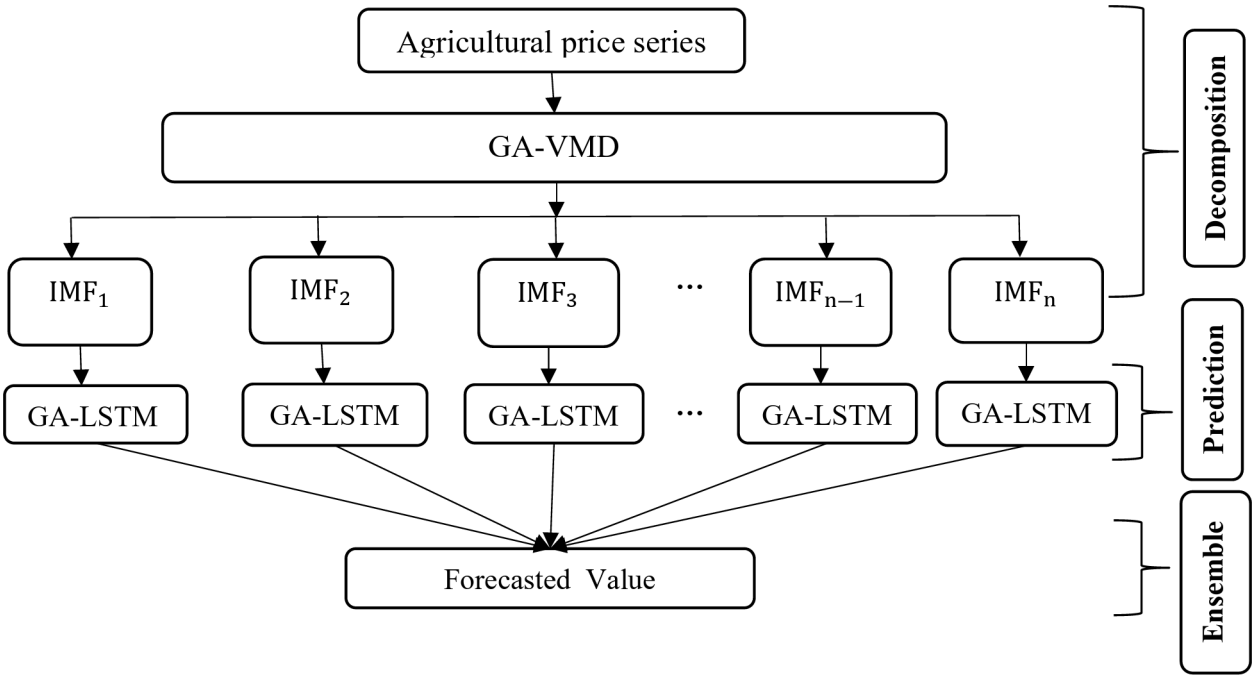


Fig. 2. Flowchart of the proposed genetic algorithm optimized hybrid model for agricultural price forecasting.

Genetic algorithm optimized long short-term memory model

Leveraging the strengths of GA-optimized VMD and LSTM, we present a novel hybrid model for forecasting agricultural prices. This model adheres to a 'Decomposition and Ensemble' approach, as illustrated in Fig. 2. The methodology comprises three key stages:

1. **Data decomposition**: GA-based VMD decomposes a price series into a different set of IMFs. Each IMF represents oscillations of agricultural prices at different frequency levels, ranging from high to low. This decomposition simplifies the analysis by isolating distinct patterns within the data, as the IMFs exhibit a regular structure with stable fluctuations. This step makes it easier to identify and model the underlying trends and behaviors in the price data.
2. **Individual prediction**: Each decomposed IMF is divided into training and testing datasets, with a portion of the training data reserved for validation. This split ensures the forecasting model's ability to generalize to unseen data. The GA-optimized LSTM model is employed to predict each mode individually. This model is particularly effective in capturing the nonlinear relationships and long-term dependencies within the IMFs, providing accurate predictions for each frequency component.
3. **Ensemble prediction**: The final forecast of the original price series is obtained by summing up the predicted values of all Intrinsic Mode Functions (IMFs) together as: $\widehat{y}(t) = \sum_{j=1}^{n} \widehat{c}_j(t)$ where, $\sum_{j=1}^{n} \widehat{c}_j(t)$represent the ensemble of predicted values of modes.

VMD is instrumental in decomposing complex, nonstationary price series into IMFs, effectively mitigating noise and extracting essential price dynamics. Unlike existing VMD-based models that rely on manually tuned parameters, our model employs GA to optimize the selection of VMD parameters [number of modes (n), balancing parameter of the data-fidelity constrain (α), tolerance of convergence criterion (τ), and time-step of the dual ascent (ε)]. This automated optimization process enhances the decomposition's precision and adaptability to various datasets, leading to more reliable and granular representations of price series. Each decomposed IMF is individually modeled using GA-optimized LSTM networks, ensuring that each IMF's temporal dependencies and intricate patterns are captured effectively. The GA-optimized LSTM further refines hyperparameter (input nodes, hidden nodes, batch size, and number of epochs) tuning, improving the model's capacity to handle nonlinear and complex price variations. This contrasts with conventional models that either apply LSTM directly to raw data or lack IMF-specific modelling, thereby missing critical component-level insights. By aggregating IMF-specific forecasts, our model produces an ensemble output that reconstructs the original price series with enhanced accuracy and reduced forecasting errors. This ensemble approach surpasses single-model strategies by leveraging the strengths of each decomposed IMF, ultimately resulting in superior performance.

Forecasting evaluation criteria

In this paper, each forecast model is assessed based on several evaluation criteria, including root mean squared error (RMSE), mean absolute percentage error (MAPE), directional prediction statistics (D_{stat}), and the Diebold-Mariano (DM) test. These multiple evaluation criteria are utilized as individual decision criteria

might not fully capture all aspects of prediction errors[11]. Moreover, for the ranking of each model, the TOPSIS method[43] is employed, which ranks each model by giving weights after normalizing the decision matrix of all evaluation criteria and calculating the geometric distance between the different models. The following are the forecasting evaluation criteria for comparing the proposed model with other models:-.

$$RMSE = \sqrt{\frac{\sum_{t=1}^{h} (y(t) - \widehat{y}(t))^2}{h}} \tag{1}$$

$$MAPE = \frac{1}{h}\sum_{t=1}^{h}\left|\frac{y(t) - \widehat{y}(t)}{y(t)}\right| \tag{2}$$

$$D_{stat} = \frac{1}{h}\sum\nolimits_{t=1}^{h} a_t \times 100\% \tag{3}$$

where $y(t)$ and $\widehat{y}(t)$ are the actual value and predicted value, respectively, h is the size of the testing set, and $a_t = \begin{cases} 1, & if\ [y(t+1) - y(t)]\,[\widehat{y}(t+1) - y(t)] \geq 0 \\ 0, & otherwise \end{cases}$

Diebold-mariano test

For a given time series $y(t)$, the Diebold-Mariano (DM) test statistics is defined as:

$$z_{DM} = \frac{\bar{d}}{\sqrt{\widehat{V}_{\bar{d}}}}$$

where h is the test size, $\{e_{tet}\}_{t=1}^{h}$ and $\{e_{ref}\}_{t=1}^{h}$ are errors for the test model and reference model, respectively, g is the loss function, $\bar{d} = \frac{1}{h}\sum_{t=1}^{h}[g(e_{tet}) - g(e_{ref})]$ is the sample mean, $\widehat{V}_{\bar{d}} = \frac{1}{h}\left[\gamma_0 + 2\sum_{j=1}^{l-1}\gamma_j\right]$ is the estimate of variance using l-step forecasts and $\gamma_j = cov(d_t, d_{t-j})$ is the estimate of jth autocovariance of $[g(e_{tet}) - g(e_{ref})]$.

Technique for order of preference by similarity to ideal solution

For a given decision matrix $X = (x_{ij})$ and a weight vector $W = [w_1, w_2, \cdots, w_e]$, rank of ith model is defined as:

$$R_i = \frac{d_i^-}{d_i^- + d_i^+}$$

where x_{ij} denotes j^{th} evaluation criteria for i^{th} prediction model for $1 \leq i \leq m$ and $1 \leq j \leq e$,, $d_i^+ = \sqrt{\sum_{j=1}^{e}(v_{ij} - v_j^+)^2}$ and $d_i^- = \sqrt{\sum_{j=1}^{e}(v_{ij} - v_j^-)^2}$ are measures of separation between the positive and negative ideal solutions, $v_{ij} = w_j * n_{ij}$ is a weighted normalized decision matrix where $\sum_{j=1}^{e} w_j = 1$, $n_{ij} = \frac{x_{ij}}{\sqrt{\sum_i x_{ij}^2}}$ is the normalized value of x_{ij}, $v_j^+ = \begin{cases} \max v_{ij}, if\ j\ is\ a\ positive\ criterion \\ \min v_{ij}, if\ j\ is\ a\ negative\ criterion \end{cases}$ $and\ \ v_j^- = \begin{cases} \min v_{ij}, if\ j\ is\ a\ positive\ criterion \\ \max v_{ij}, if\ j\ is\ a\ negative\ criterion \end{cases}$

are extremely positive and extremely negative performance on each criterion

Results and discussion

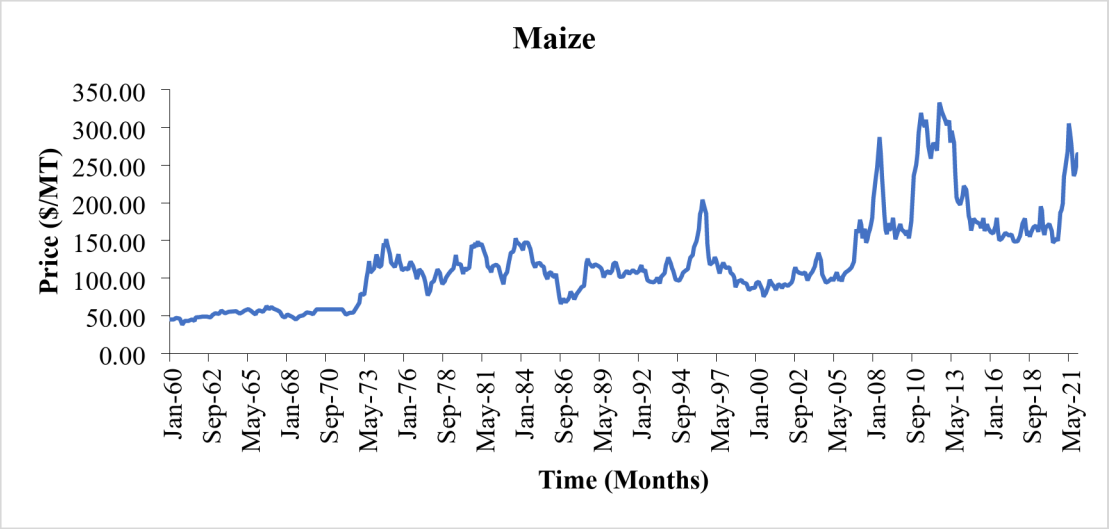
Data set

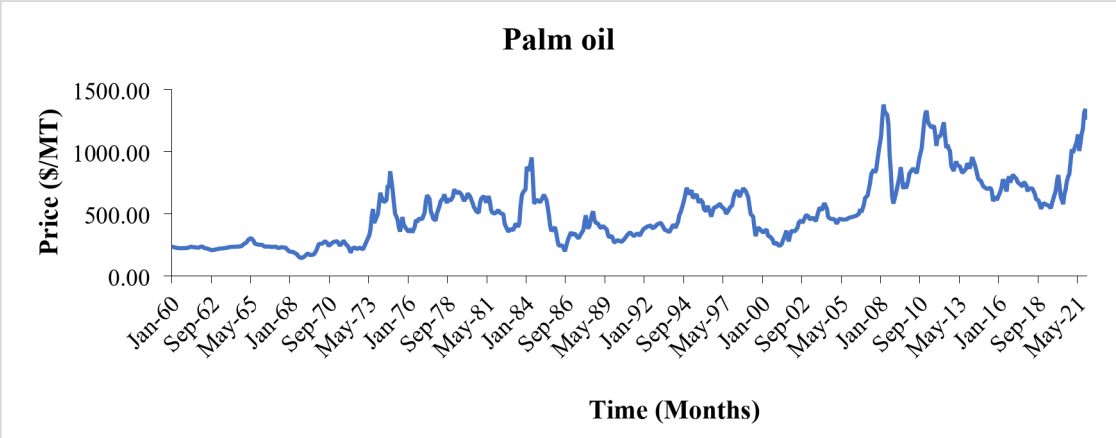
This study utilizes data from three monthly international price series: Maize, Palm oil, and Soybean oil (measured in dollars per metric tonne, $/MT). The data is sourced from the "World Bank Commodity Prices Data" (https://www.worldbank.org/en/research/commodity-markets) and spans from January 1960 to December 2021.

The three price series were divided into two sets for analysis. The first set consists of the training and validation sets from January 1960 to December 2020, totalling 732 observations. The second set is the testing set, comprising the remaining 12 observations. The data is visualized using time plots for each price series to begin the time series analysis, as illustrated in Fig. 3. These time plots reveal the nonlinear and nonstationary behaviour commonly observed in agricultural price data. For further insights, Table 3 presents the basic descriptive statistics for each price series.

Tests for stationarity and linearity

Since agricultural price series are complex and may exhibit nonstationarity and nonlinearity properties, it becomes necessary to test these properties, which can be helpful in the skilful handling of data while fitting the model. The Augmented Dickey-Fuller (ADF)[44] and Brock-Dechert-Scheinkman (BDS) tests[21] are used to check the stationarity and linearity characteristics, respectively.





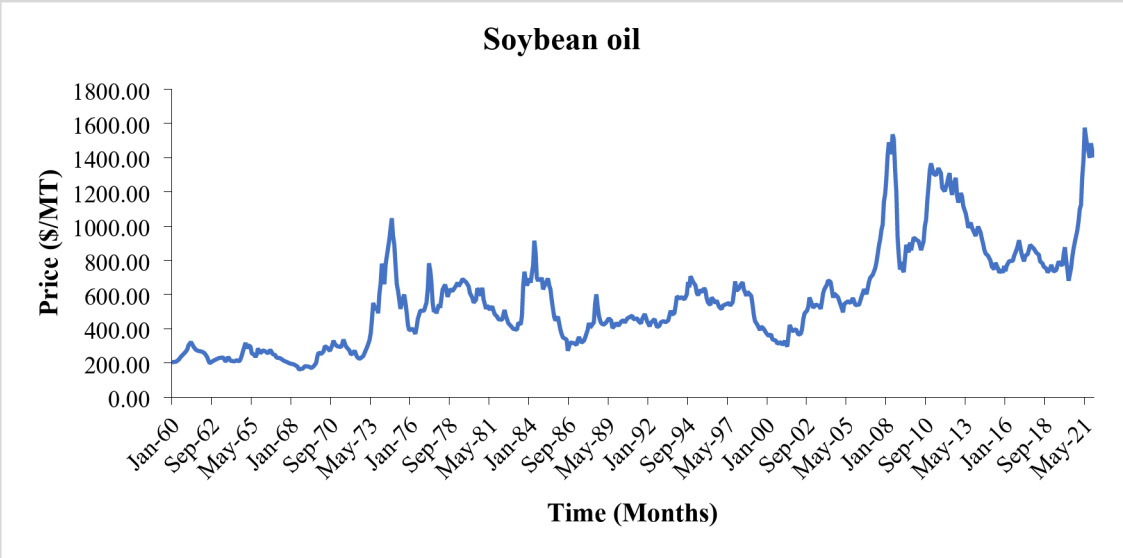


Fig. 3. Time plots for monthly international Maize, Palm oil, and Soybean oil prices ($/MT) series.

Statistics	Maize	Palm oil	Soybean oil
Mean	120.31	512.49	574.07
Maximum	333.05	1377.22	1574.67
Minimum	38.00	141.73	157.00
Standard deviation	59.65	257.75	294.08
Skewness	1.28	0.93	1.01
Kurtosis	1.80	3.58	0.82
Jarque-Bera	299.40	118.26	146.97

Table 3. Descriptive statistics of the price ($/MT) series.

Series	Augmented Dickey-Fuller Test		Conclusion
	t-statistic	Probability	
Maize	-3.12	0.10	Nonstationary
Palm oil	-3.14	0.09	Nonstationary
Soybean oil	-3.01	0.15	Nonstationary

Table 4. Augmented Dickey-Fuller test results.

Series	Embedding dimension					Conclusion
		2		3		
	Epsilon	Statistics	Probability	Statistics	Probability	
Maize	0.5 σ	133.23	< 0.001	224.66	< 0.001	Nonlinear
	1.0 σ	67.16	< 0.001	77.15	< 0.001	
	1.5 σ	52.64	< 0.001	53.64	< 0.001	
	2.0 σ	41.41	< 0.001	39.72	< 0.001	
Palm oil	0.5 σ	230.22	< 0.001	399.29	< 0.001	Nonlinear
	1.0 σ	98.24	< 0.001	119.26	< 0.001	
	1.5 σ	64.78	< 0.001	67.36	< 0.001	
	2.0 σ	53.68	< 0.001	51.77	< 0.001	
Soybean oil	0.5 σ	193.52	< 0.001	335.36	< 0.001	Nonlinear
	1.0 σ	85.31	< 0.001	102.54	< 0.001	
	1.5 σ	60.46	< 0.001	62.45	< 0.001	
	2.0 σ	52.53	< 0.001	50.74	< 0.001	

Table 5. Brock-Dechert-Scheinkman test results.

Table 4 shows the ADF test results confirming the nonstationarity of each price series. The test result demonstrates that the price series of maize, palm oil, and soybean oil are nonstationary at 10%, 9%, and 15% significance, respectively. The BDS test is conducted to assess the spatial dependence of a time series. Thus, to assess nonlinearity, we conducted a BDS test with embedding dimensions of 2 and 3, a time delay of 4, and a distance threshold of 0.5 standard deviations. The results, summarized in Table 5, revealed statistically significant nonlinearity for all price series. Specifically, the probability values associated with each series were consistently below 0.001 for both embedding dimensions, strongly rejecting the null hypothesis of linearity at the 1% significance level. The significance of nonlinearity demonstrates agricultural price movements' intricate and unpredictable nature. This complexity emphasizes the importance of decomposition methods, which break down the series into simpler, more stable components, enabling more accurate modeling of each component and ultimately improving forecasting accuracy.

Decomposition of the agricultural price series

After checking prerequisite conditions for decomposition, i.e. nonlinearity and nonstationarity characteristic of agricultural price series, we use four decomposition tools: EMD, EEMD, CEEMDAN, and VMD. These tools effectively break down the original price series into simpler and more stable components for better analysis. Techniques like Empirical Mode Decomposition (EMD) and its variants—Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN)—decompose the series into a set of Intrinsic Mode Functions (IMFs) and a residual. The total number of modes (IMFs and residual) is approximately $\log_2 T$, where T is the total data points. Besides the

Parameter	Value
Chromosome representation	Bit Strings
Population size	100
Generations	25
Mutation rate	0.1
Crossover rate	0.5
Fitness function	r_{res} (VMD) and RMSE (LSTM)

Table 6. Parameters utilized by genetic algorithm for each price series to get optimum results.

Series	α	τ	*n*	ε	r_{res}
Maize	1000	1	12	1.00E-07	0.15
Palm oil	1000	0	13	1.00E-07	0.12
Soybean oil	1000	1	12	1.00E-06	0.10

Table 7. Variational mode decomposition parameters through genetic algorithm for different price series. α is balancing parameter of the data-fidelity constraint , τ $is\ tolerance\ of\ convergence\ criterion,$ n is number of modes, ϵ is time-step of the dual ascent, and r_{res} is ratio of residual energy.

number of modes, the EMD requires two more hyperparameters, i.e., *the* stopping criterion and the maximum number of siftings. EEMD and CEEMDAN need four additional hyperparameters: ensemble size (), noise strength, stopping criterion, and maximum number of siftings. These parameters are optimized through experimentation to minimize the energy evaluation parameter (), ensuring superior decomposition. For EEMD and CEEMDAN, commonly used values include an ensemble size of 250, noise strength at 0.2 times the standard deviation of the series, a stopping criterion of 4, and a maximum of 50 siftings.

VMD relies on four key hyperparameters that significantly influence the decomposition process and structure. These include the number of modes (n), which determines how many components the data is decomposed into; the balancing parameter of the data-fidelity constraint (α), which controls the trade-off between the data's fidelity and smoothness of the modes; the tolerance of the convergence criterion (τ), which ensures the decomposition process stops once the solution stabilizes; and the time-step of the dual ascent (ε), which governs the rate of update during optimization. Selecting optimal values for these parameters is critical to achieving effective and accurate decomposition tailored to the specific characteristics of the given dataset. While doing this, some theoretical considerations must be considered for each hyperparameter so that proper search space can be employed and optimum combinations can be obtained in less time. For example, If the number of modes is too small, it may not be sufficient to decompose the original series, resulting in inaccurate predictions adequately. Conversely, if the time series is excessively decomposed, the differences between each component become minimal, leading to reduced accuracy and unnecessary computing overheads. Since the grid search approach to generate the optimum model may be exceedingly time-consuming and tedious, this task can be automated using genetic algorithms that optimize results by selecting candidates from their search space in a random manner and achieving results in less time. We apply binary-coded GA to our model. The hyperparameter values range used in this study are as follows: (i) number of modes: $\boldsymbol{n} \in \{2,3,4 \cdots 25\}$ (ii) balancing parameter of the data-fidelity constrain : $\boldsymbol{\alpha} \in \left\{10, 10^2, 10^3, 10^4, 10^5\right\}$ (iii) tolerance of convergence criterion: $\boldsymbol{\tau} \in \{0,1\}$ and (iv) time-step of the dual ascent: $\boldsymbol{\epsilon} \in \left\{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\right\}$. The fitness function for GA is written as follows:

$$\text{Ratio of residual energy}(r_{res}) = \frac{\sum_{t=1}^{T} \left| \frac{y(t) - \sum_{j=1}^{n} c_j(t)}{y(t)} \right|}{T}$$

Moreover, besides determining the fitness functions and problem representation, GA has some other controlling parameters. Table 6 lists the GA parameters in detail. When parameters for VMD are set according to Table 7, the VMD-Loss is the minimum for all three agricultural commodities. For the above parameters, residual error after decomposition (r_{res}) is used as a fitness function, as explained in the methodology section. Empirically, when r_{res} is less than 1% and there is no significant downward trend, the final VMD parameters can be determined. As shown in Table 7, all r_{res} are below 1%. Finally, GA-optimized VMD, gives the number of modes as 12,13, and 12 for maize, palm oil, and soybean oil, respectively. These extracted subsequences are smoother and more regular than the original data, which is conducive to improving forecasting accuracy. Figure 4 shows the three price series' decomposed modes through GA-based VMD from high frequency to low frequency. Here, high frequency shows the effect of short-term fluctuations in the market.

In contrast, low frequency represents any particularly significant event affecting the demand-supply equilibrium at that time. For instance, in our case, the two most significant events were observed in 2008 and 2011 which can be observed in Fig. 4 shows spikes around the 580th and 620th observations, respectively. Both events can be attributed to the 2007-08 global food crisis and the rise in biofuel production[45]. The utilization

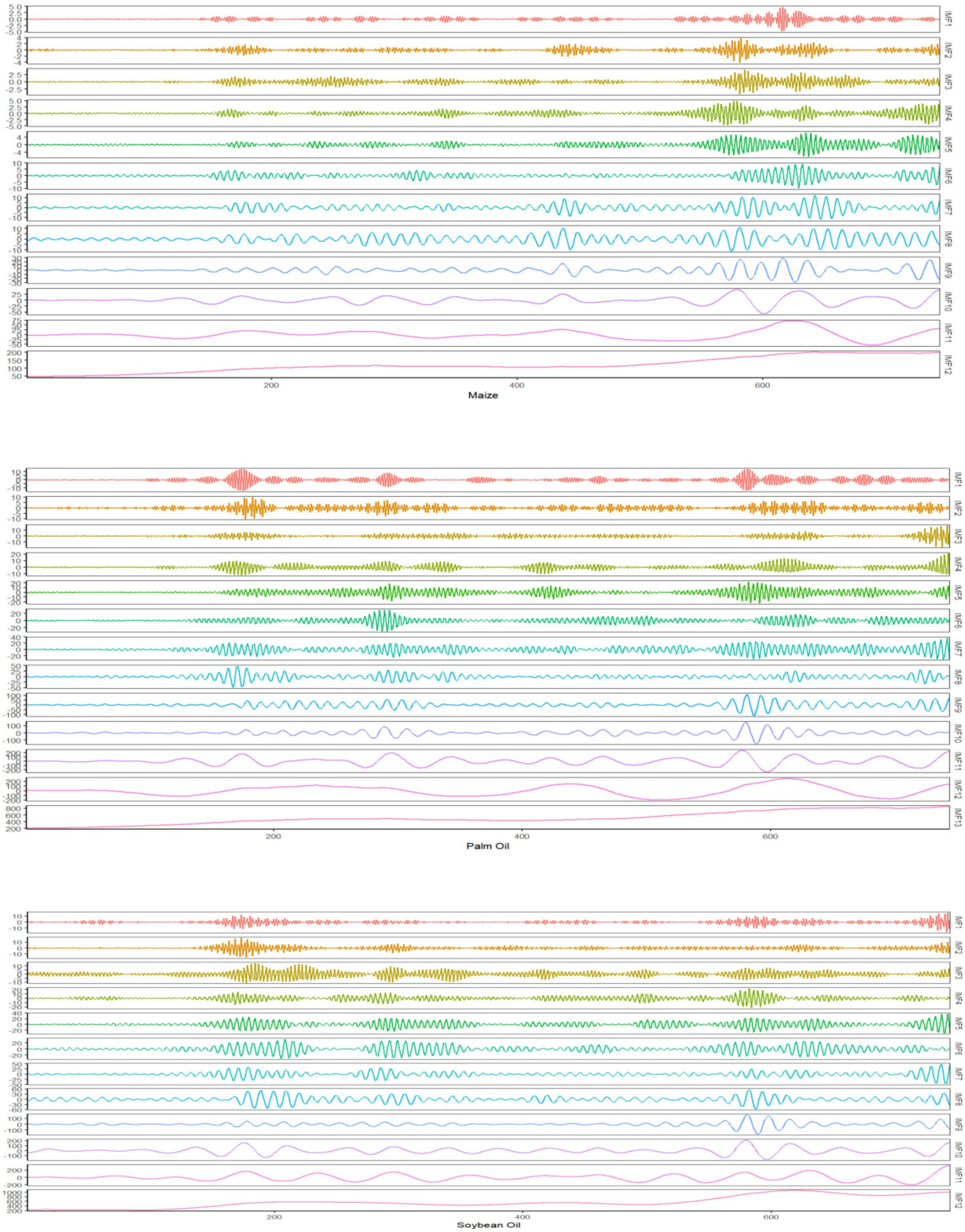


Fig. 4. The decomposed IMFs for Maize, Palm oil and Soybean oil price series through genetic algorithm optimized variational mode decomposition.

Series	EMD	EEMD	CEEMDAN	VMD
Maize	0.1669	0.0472	0.0435	0.0214
Palm oil	0.0420	0.0292	0.0190	0.0121
Soybean oil	0.0608	0.0443	0.0279	0.0140

Table 8. Comparison of different decomposition algorithm in terms of energy evaluation parameter ($\boldsymbol{\theta}$)

Series	Number of Input Nodes	Number of Hidden Nodes	Batch Size	Epoch
Maize	4	6	64	60
Palm oil	3	8	32	60
Soybean oil	5	12	64	40

Table 9. Model hyperparameters utilized by the genetic algorithm optimized long short-term memory model for different price series.

of maize for ethanol fuel escalated from 15% (in 2006) to 40% (in 2012) of the entire U.S. maize production. Furthermore, the Intrinsic Mode Functions (IMFs) obtained from VMD-based decomposition exhibit a more independent and distinct frequency distribution compared to those derived from EMD variants. This enhanced independence can be empirically validated by analyzing the energy evaluation parameter (θ)[46], defined as:

$$\theta = \frac{\left|\sqrt{\sum_{j=1}^{n} E_{j(t)}^2} - E_{y(t)}\right|}{E_{y(t)}};$$

$$E_{y(t)} = \sqrt{\frac{\sum_{t=1}^{T} y^2(t)}{T}}$$

where $E_{y(t)}$ and $E_{j(t)}$ are the energy values of the original time series and j^{th}IMF, respectively. In this context, is used as an evaluation parameter to assess the orthogonality of the Intrinsic Mode Functions (IMFs). A value of closer to 0 signifies a higher degree of orthogonality between the IMFs, indicating that the components are well-separated and distinct. Conversely, a higher value of suggests the presence of overlapping or elusive components within the IMFs, implying that the decomposition may not be as effective in isolating independent frequency modes. Table 8 compares different decomposition methods in terms of θ for each price series, which shows that the value of θ in the case of VMD is the smallest.

Forecasting results

The forecasting performance of the GA-optimized VMD-LSTM model is evaluated by comparing it with four other models: LSTM, EMD-LSTM, EEMD-LSTM, and CEEMDAN-LSTM. To facilitate this comparison, we developed two R software packages, named decompDL[38] and TSdeeplearning[39], which are publicly available on the Comprehensive R Archive Network (CRAN). These packages were specifically designed to support the decomposition and deep learning tasks required for time series forecasting, providing a robust platform for model implementation and evaluation. The comparison is based on international monthly maize, palm, and soybean oil prices.

A standard LSTM model is developed to forecast each price series individually in this comparison. Whereas under the hybrid model (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM, and proposed VMD-LSTM), the LSTM network is employed to predict each mode. In this paper, we utilized Keras to construct an LSTM network. The hyperparameter values range used in this paper for GA optimization are as follows: (1) input nodes: 1–15 and (ii) hidden nodes: 1–15 (iii) batch size: 20–70 (iv) number of epochs: 30–100. Table 9 displays all the hyperparameters used in our analysis for individual LSTM model for each price series. In addition to evaluating the forecasting performance, we also assessed the learning performance of the optimized LSTM model to ensure that it is a good fit without overfitting or underfitting. Epoch-wise learning curves for the training and validation sets were generated to visualize the learning performance. Figure 5 exhibits the epoch-wise learning curves acquired from the optimized models for the respective time series. The plots demonstrate a strong alignment between the training and validation curves in all three cases, signifying the outstanding learning performance of the optimized model. After finalizing the hyperparameters, a standard LSTM model is applied to train the training set of each price series. Figure 6 illustrates the model's fitted values and predicted errors for the training set. The figure clearly demonstrates that the prediction errors of the LSTM model are symmetrically distributed around zero, with relatively small fluctuations. This pattern strongly indicates the superior predictive performance and flexibility of the LSTM model, especially when applied to a variety of datasets. In a similar manner, each Intrinsic Mode Function (IMF) is predicted individually using the LSTM, which effectively captures the specific patterns within each IMF. Ultimately, the forecast for the original price series is obtained by linearly summing the predictions of all the IMFs, providing a comprehensive and accurate forecast for the entire series. Figure 7 demonstrates the forecasting results of each predicted model for the three agriculture price series in the test

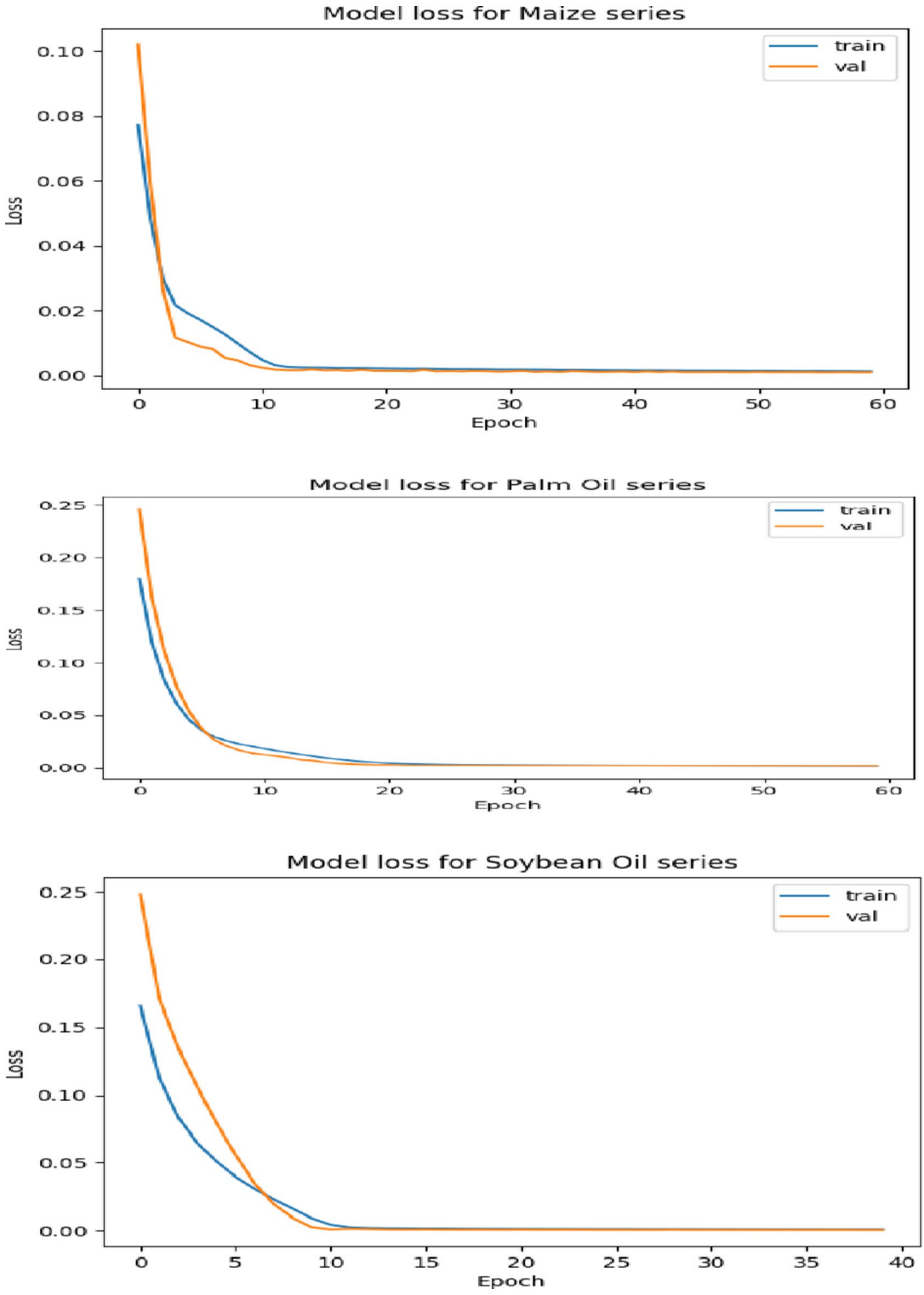
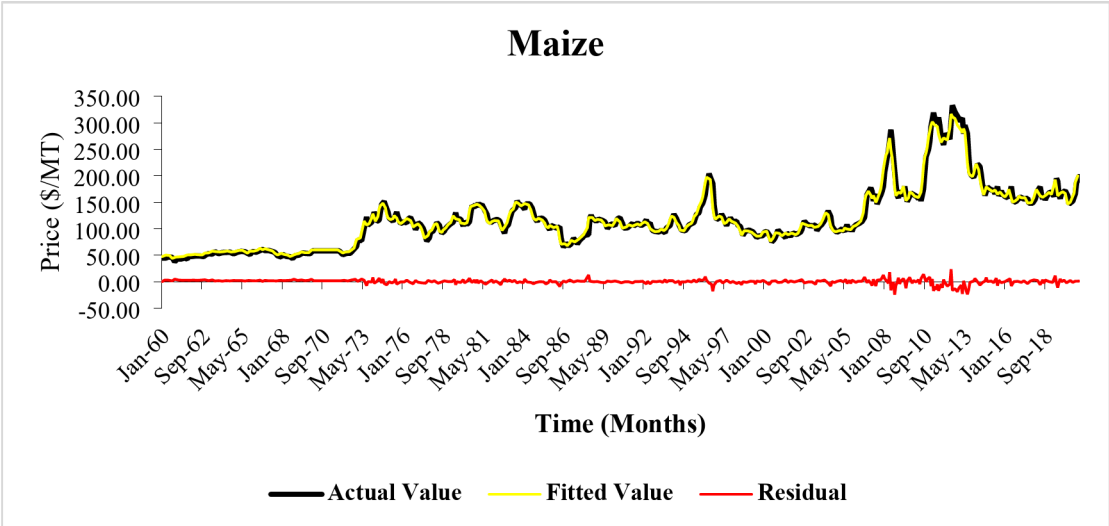
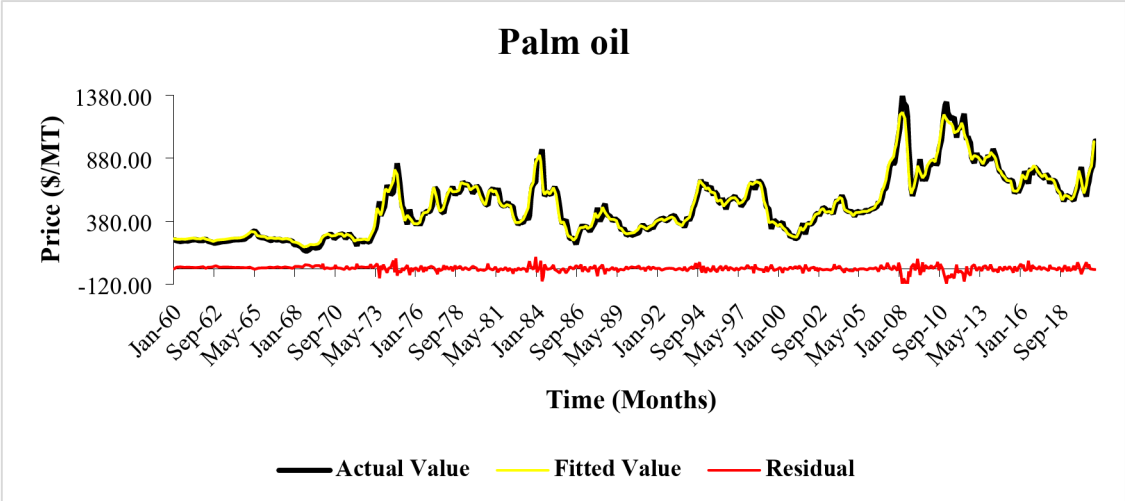


Fig. 5. Learning curves of the genetic algorithm optimized long short-term memory model for training and validation sets.

set. Table 10 quantitatively and Fig. 8 visually display a detailed comparison of the MAPE, RMSE, D_{stat} and TOPSIS for each prediction model. Table 11 displays the DM test results among individual models (LSTM) and its decomposition-based hybrid models (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM, and VMD-LSTM) for each price series. The following can be concluded from these tables and figures:





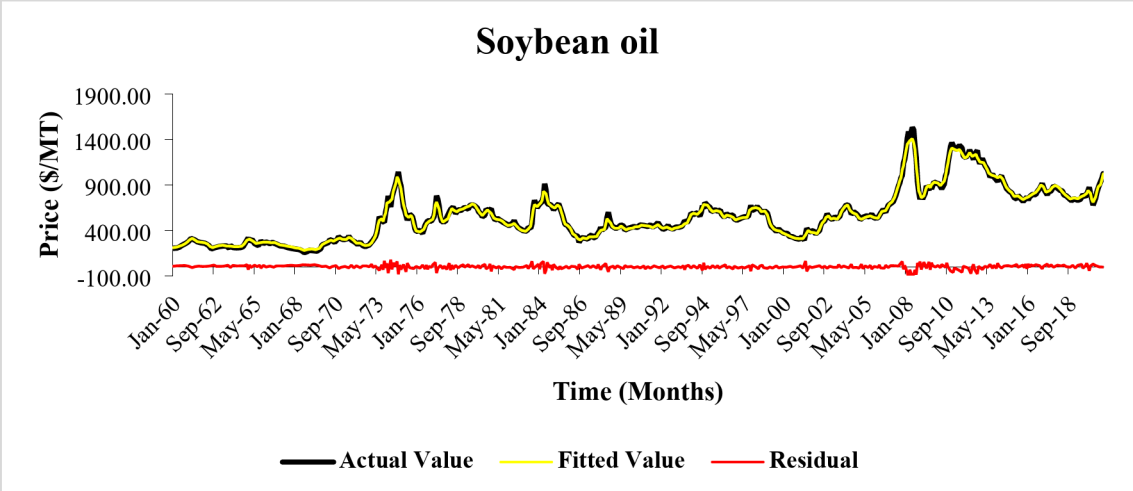
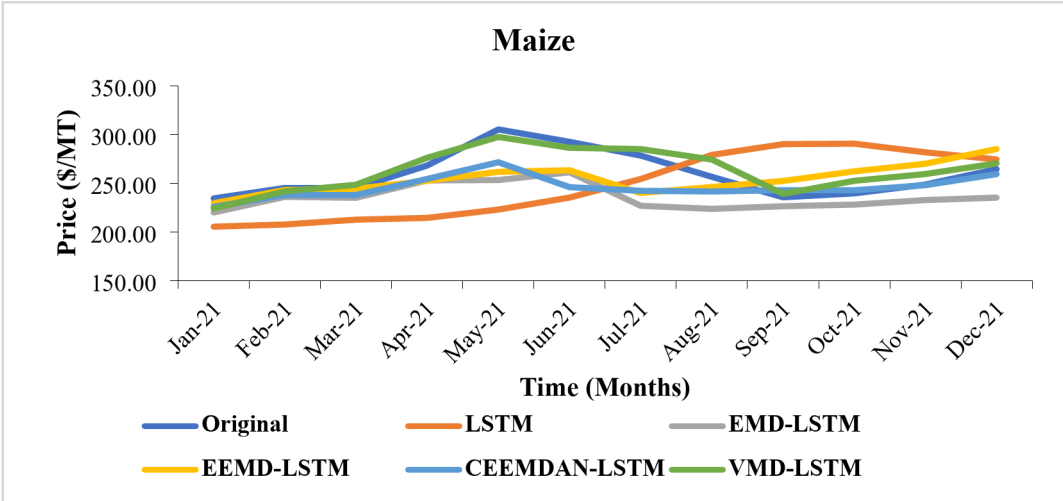
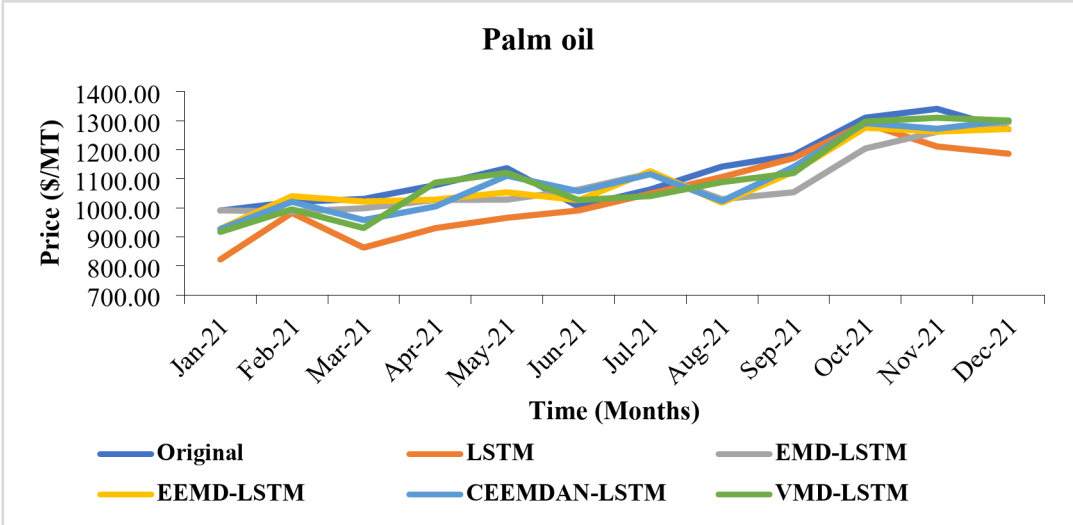


Fig. 6. The fitted value with observed series and residual of maize, palm oil, and soybean oil price for training set from genetic algorithm optimized long short-term memory model.





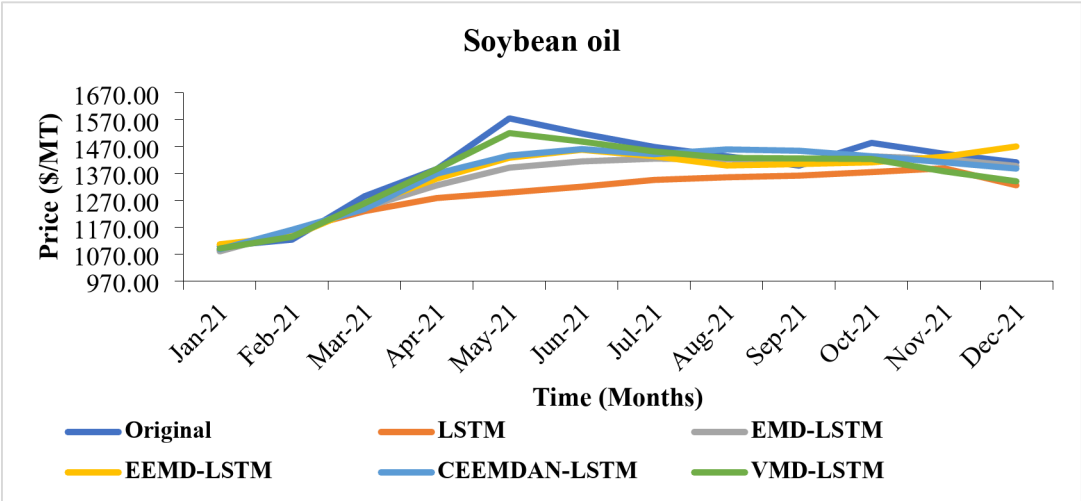


Fig. 7. Predicted results of different models.

1) The "decomposition-and-ensemble" hybrid models (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM, and VMD-LSTM) consistently outperform the standalone LSTM by decomposing complex price series into simpler, more stable IMFs. The decomposition mitigates noise and isolates distinct patterns within the data, allowing the LSTM to model and predict each component more effectively. In contrast, the standalone LSTM

Forecasting Models	Maize			Palm oil			Soybean oil			TOPSIS Rank
	MAPE	RMSE	D_{stat}	MAPE	RMSE	D_{stat}	MAPE	RMSE	D_{stat}	
LSTM	0.1553	44.81	54.54	0.0752	106.11	36.36	0.0671	121.06	72.73	5
EMD-LSTM	0.0873	27.93	63.64	0.0569	76.40	54.54	0.0342	68.56	100	4
EEMD-LSTM	0.0701	22.87	90.90	0.0445	60.78	54.54	0.0304	57.03	90.90	3
CEEMDAN-LSTM	0.0559	20.99	100	0.0443	59.63	54.54	0.0294	53.96	90.90	2
VMD-LSTM	0.0313	9.04	90.90	0.0347	46.61	72.73	0.0218	39.39	100	1

Table 10. Forecasting performance of different models for maize, palm oil, and soybean oil price. *MAPE is mean absolute percentage error, RMSE is root mean squared error,* D_{stat} *is directional prediction statistics, and TOPSIS is technique for order of preference by similarity to ideal solution.*

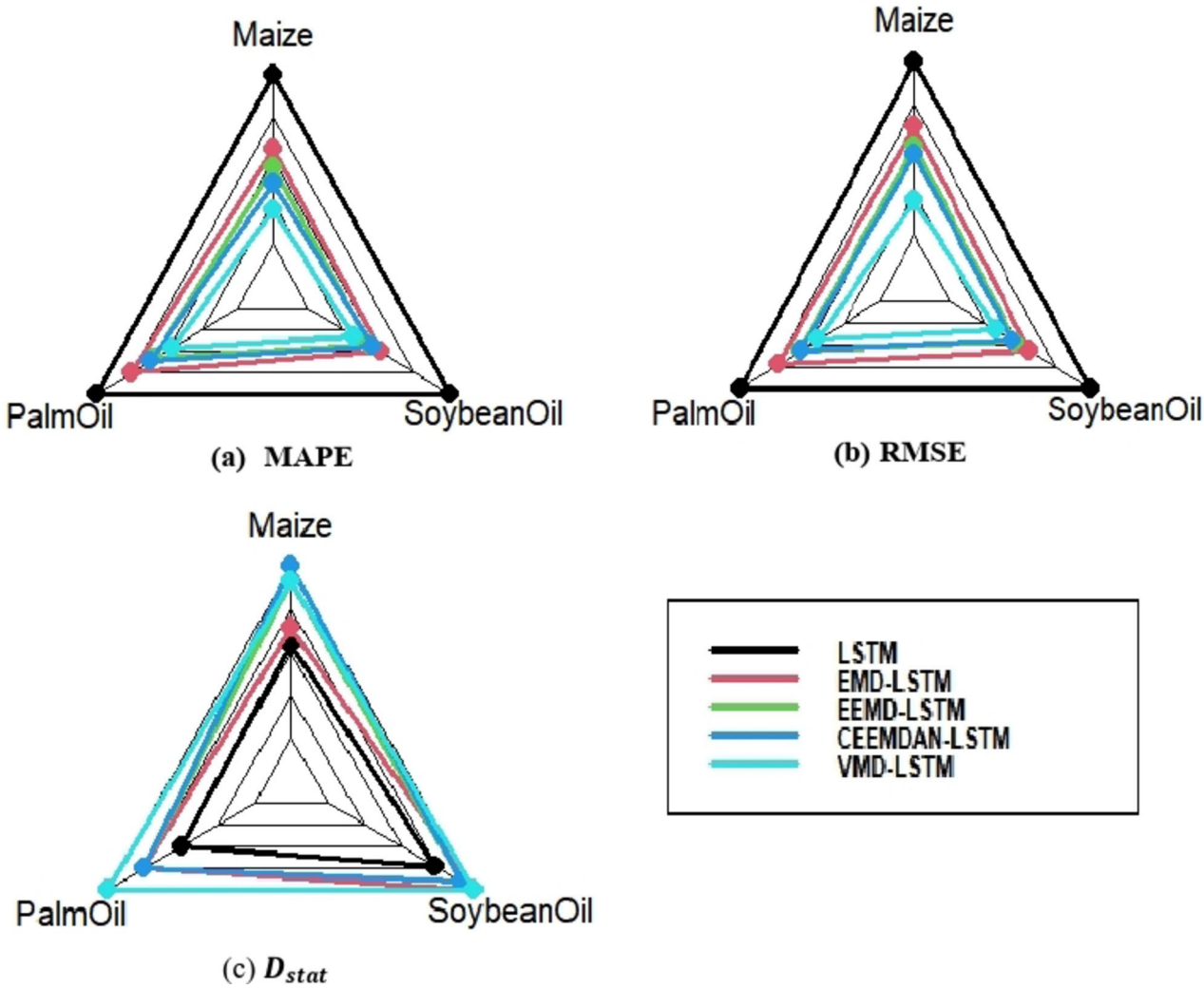


Fig. 8. Comparative radar charts of different forecasting evaluation criteria for Maize, Palm oil, and Soybean oil price series.

directly processes the raw price series, making it more vulnerable to volatility and fluctuations, often resulting in higher forecasting errors. According to Table 10, the VMD-LSTM model consistently achieves the lowest MAPE and RMSE values, indicating superior accuracy in level forecasting. For maize, the MAPE reduces from 0.1553 in the LSTM model to 0.0313 in the VMD-LSTM model, representing a reduction of 79.84%. Similarly, the RMSE decreases by 79.83%, from 44.81 to 9.04. A similar trend is observed for palm oil and soybean oil, where the MAPE decreases by 53.85% and 67.51%, respectively, and the RMSE decreases by 56.07% and 67.46%, respectively. The directional accuracy also improves with the application of decomposition-based models. The D_{stat} for maize increases from 54.54% in the LSTM model to 90.90% in the

Series	Tested Model	Benchmark Models			
		LSTM	EMD-LSTM	EEMD-LSTM	CEEMDAN-LSTM
Maize	EMD-LSTM	2.49(0.015)			
	EEMD-LSTM	4.08(0.000)	1.69(0.059)		
	CEEMDAN-LSTM	4.48(0.000)	2.64(0.011)	1.03(0.161)	
	VMD-LSTM	5.40(0.000)	3.42(0.002)	2.66(0.010)	1.58(0.070)
Palm oil	EMD-LSTM	2.66(0.010)			
	EEMD-LSTM	3.50(0.002)	0.99(0.170)		
	CEEMDAN-LSTM	3.58(0.001)	1.46(0.085)	2.17(0.021)	
	VMD-LSTM	4.36(0.000)	1.76(0.052)	2.34(0.019)	2.36(0.018)
Soybean oil	EMD-LSTM	4.86(0.000)			
	EEMD-LSTM	4.41(0.000)	0.78(0.225)		
	CEEMDAN-LSTM	3.50(0.002)	0.88(0.197)	1.03(0.161)	
	VMD-LSTM	3.21(0.004)	1.23(0.121)	1.13(0.140)	1.26(0.116)

Table 11. Forecasting performance in terms of Diebold-Mariano test of different models for maize, palm oil, and soybean oil price for the one-year forecast horizon.

VMD-LSTM model. For palm oil, D_{stat} improves from 36.36 to 72.73%, and for soybean oil, it increases from 72.73 to 100%. These results highlight the effectiveness of the VMD-LSTM model in capturing both level and directional trends in price series data. Tabel 11 (DM test) also demonstrate that all hybrid models statistically significant improvements over the single-prediction LSTM model. Specifically, at a 1% significance level, all hybrid models outperform LSTM in predicting price movements for maize, palm oil, and soybean oil. Finally, the results demonstrate that hybrid models achieve lower RMSE and MAPE values, higher D_{stat} scores, and significant DM statistics reflecting improved forecasting accuracy and better capture of both short-term fluctuations and long-term trends. The TOPSIS analysis ranks the standalone LSTM lowest, further reinforcing the advantage of decomposition-based approaches. The primary reason for decomposition improves forecasting accuracy is that agricultural price series are often characterized by sharp fluctuations, irregular patterns, and overlapping trends, making them highly volatile and complex. When a standalone LSTM model is applied directly to such data, it faces difficulty in distinguishing between meaningful patterns and random noise. This can result in inaccurate forecasts, as the model may either overfit to noise or miss critical underlying trends. Decomposition overcomes this challenge by disintegrating the original price series into simpler, more stable intrinsic mode functions (IMFs). Each IMF represents different frequency components, with high-frequency IMFs capturing short-term fluctuations and noise, while low-frequency IMFs reflect underlying trends and seasonal cycles. This decomposition allows the LSTM to model and predict each component individually, focusing on smaller, less volatile patterns. As a result, the hybrid models effectively isolate and process various elements of the price series, reducing overall forecasting error. When the predictions of all IMFs are aggregated, the final forecast accurately reflects the complex dynamics of the original price series, leading to significant improvements in both level and directional prediction accuracy. This decomposition approach enhances the model's capacity to handle nonstationary and nonlinear price data, resulting in more precise and reliable predictions.

2) Among decomposition-based models, VMD-LSTM outperforms the EMD and its variants-based LSTM models. As Table 10 shows, the level forecasting measure MAPE reduces up to 19.70%, 21.80%, and 11.12%; and RMSE reduces up to 18.12%, 20.45%, and 16.82% when we use EEMD-LSTM as a forecasting model instead of EMD-LSTM for maize, palm oil, and soybean oil, respectively. When we use CEEMDAN as an advanced decomposition tool among EMD variants with the LSTM model, i.e. CEEMDAN-LSTM, then there is again a reduction in level measure MAPE (20.25%, 1%, 3.28%) and RMSE (8.22%, 1.89%, 5.38%) in comparison to EEMD-LSTM for maize, palm oil, and soybean oil, respectively. Finally, for the proposed VMD-LSTM, the reduction in MAPE (44%, 21.67%, 25.85%) and RMSE (56.93%, 21.83%, 27.00%) compared to CEEMDAN-LSTM. Table 10 also shows that the directional prediction statistics D_{stat} observed maximum for maize through the CEEMDAN-LSTM model, palm oil through VMD-LSTM, and soybean oil through EEMD-LSTM and VMD-LSTM. Among hybrid models, TOPSIS ranks first to the proposed VMD-LSTM and fourth to EMD-LSTM. Table 11 shows that among hybrid models, VMD-LSTM performs better than all EMD-based LSTM models at the significance level of less than 1% for maize, same for palm oil except for EMD-LSTM is significant at 5%.

3) The superior performance of VMD-LSTM over EMD-based hybrid models can be attributed to several factors. Firstly, VMD is less sensitive to noisy data than the EMD decomposition method. This means that VMD can handle noisy data more effectively, leading to more accurate predictions. Secondly, VMD has better decomposition adaptability and minimum decomposition loss. This enables VMD to provide deeper decomposition subseries, which can lead to more accurate predictions. On the other hand, EMD is limited by model mixing, noise sensitivity, boundary effects, etc., which can lead to less accurate predictions. Thirdly, compared to EMD, VMD has the advantage of generating more Intrinsic Mode Functions (IMFs). This abundance of IMFs enables VMD to portray the characteristics of each component more comprehensively. Consequently, VMD is better equipped to capture the underlying patterns and trends present in the data,

resulting in more accurate predictions. Overall, VMD overcomes the limitations of EMD and has a better decomposition effect, resulting in a more remarkable improvement in prediction accuracy. Therefore, the use of VMD-LSTM over EMD-based hybrid models can lead to more accurate and reliable predictions, especially in the presence of complex agriculture price data.

Conclusions

Agriculture commodity prices play a critical role for all stakeholders involved in the agriculture industry, and price fluctuations can have far-reaching impacts on the national economy. To improve the forecasting accuracy of agriculture price, this paper proposed a novel GA-optimized VMD-LSTM forecasting model. The model utilizes the GA-optimized VMD technique, a powerful decomposition method that decomposes the original price series into multiple subseries. This decomposition reduces the volatility of the original price data, facilitating more effective analysis and forecasting. Subsequently, a GA-based LSTM network is utilized to create individual forecasting models for each independent mode obtained from the decomposition. These individual forecasts are then combined to produce an ensemble result. To assess the performance of the proposed model, experiments are conducted using monthly price data for maize, palm oil, and soybean oil from the international commodity market. The forecasting results of the proposed VMD-LSTM model are compared with LSTM, EMD-LSTM, EEMD-LSTM, and CEEMDAN-LSTM in terms of MAPE, RMSE, D_{Stat}, TOPSIS, and DM test. After a detailed study, the following conclusions can be drawn. Firstly, the decomposition-based hybrid models outperform the individual model. Secondly, among hybrid models, the proposed VMD-LSTM hybrid model performs better than its competing hybrid models. Overall, the proposed VMD-LSTM hybrid model presents a promising approach to forecasting agriculture prices, which could provide valuable insights for farmers, traders, and policymakers and could lead to more informed decision-making and a more stable agriculture industry.

Data availability

The datasets utilized and/or analysed in the present study are accessible from the corresponding author upon reasonable request.

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Author contributions
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